

Annuities

Problem 1. An annuity-immediate provides annual payments of 10 for 20 years. Immediately following the 11th payment, the annuity is exchanged for a perpetuity-immediate of equal value with semiannual payments. The present values at the time of exchange are based on an annual effective interest rate of 6%.

The first payment of the perpetuity is K , and each subsequent payment is 0.5% larger than the previous payment.

Calculate K .

Problem 2. A loan of \$60,000 is repaid over ten years with semiannual payments at a nominal interest rate of 6% compounded semiannually.

The first payment is \$3,000 and is paid 6 months from the date of the loan. For the first five years, each subsequent payment will be \$300 more than the previous payment. After the 10th payment, each succeeding semiannual payment will be reduced by X until the loan is completely paid.

Calculate X .

Problem 3. A series of payments of 1 is made at times

$$t = 10, 14, 18, \dots, 34.$$

Let i denote the annual effective interest. Find an expression of the present value at time 0 of this payment stream as a function of i .

- (A) $\frac{a_{\overline{34}|i} - a_{\overline{6}|i}}{s_{\overline{4}|i}}$
- (B) $\frac{a_{\overline{35}|i} - a_{\overline{7}|i}}{s_{\overline{4}|i}}$
- (C) $\frac{\ddot{a}_{\overline{34}|i} - \ddot{a}_{\overline{6}|i}}{s_{\overline{4}|i}}$
- (D) $\frac{a_{\overline{34}|i} - a_{\overline{6}|i}}{\ddot{s}_{\overline{4}|i}}$
- (E) $\frac{\ddot{a}_{\overline{34}|i} - \ddot{a}_{\overline{6}|i}}{\ddot{s}_{\overline{4}|i}}$
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Problem 4. A perpetuity makes continuous payments. At time t , the perpetuity pays at an annual rate of $3t$, where time is measured in years.

At an annual effective interest rate of 5%, calculate the present value of the perpetuity.

Problem 5. Jen invests $\$X$ at the beginning of each year for 6 years at an annual effective interest rate of 5%. The interest credited at the end of each year is reinvested at an annual effective interest rate of 7%.

Bill invests $\$X$ at the end of each year for 10 years at an annual effective interest rate of $i\%$. The accumulated value of Bill's investments at the end of 10 years is twice as much as the accumulated value of Jen's investments at the end of 6 years.

Calculate i .

Problem A. \$1,000 is deposited into a savings account at time $t = 0$. No other amounts are deposited. The accumulated amount in the account at any time is given by

$$A(t) = 1000 \left(1 + \frac{2t}{35} \right)^2.$$

At what time t is the force of interest equal to 0.04?

Problem B. Joe deposits \$10 today and another \$30 in five years into a fund paying simple interest of 11% per year.

Tina will make the same two deposits, but the \$10 will be deposited n years from today and the \$30 will be deposited $2n$ years from today. Tina's deposits earn an annual effective interest rate of 9.15%.

At the end of 10 years, the accumulated amount of Tina's deposits equals the accumulated amount of Joe's deposits.

Calculate n .

Problem C. Jim deposits \$100 into an account that credits:

- An effective rate of discount of 3% for the first year;
- An effective rate of discount of 6% for the second year;
- An effective rate of discount of 9% for the third year; and
- $\delta_t = 0.02t$ for the fourth and fifth years.

How much is in the account at the end of the fifth year?
